

KRCM / CBHO

Complete Unified Working Record

Tests 1 through 54 in one continuous sequence

Kinetic Recoil Cosmological Model / Cosmic Black Hole Object

Compiled 22 September 2026

Working-status note. This is a speculative, non-peer-reviewed research record. PASS means a stated mathematical or numerical gate passed inside the declared model and assumptions. It does not mean the physical hypothesis is established. Reduced, proxy, local, historical, and conditional results are labeled explicitly.

Contents

1. Executive summary
 2. Unified theory statement and frozen Trial Model 1
 3. Unified authoritative test sequence - Tests 1 through 54
 4. Detailed record - Tests 1 through 16
 5. Detailed record - Tests 17 through 30
 6. Detailed record - Tests 31 through 42
 7. Detailed record - Tests 43 through 54
 8. Other retained numerical and conceptual work
 9. Failures, invalid approaches, and corrections that remain part of the record
 10. Current scientific status and next gates
- Appendix A. Historical numbering crosswalk for archived Tests 18-22
- Appendix B. Public critique / Reddit posting guidance

1. Executive summary

The current strongest internal result is Test 52: the frozen Trial Model 1 admits a constraint-compatible, monotone single-throat radial slice connecting the retained daughter-neck state to the correct expanding parent vacuum. Test 53 evolves that slice nonlinearly for a short interval and retains a regular throat with approximately second-order constraint convergence. Test 54 extends the evolution and finds a bifurcation into a central bulge plus two symmetric off-center throats rather than immediate pinch-off. These remain model-internal numerical results, not observational proof of the cosmological hypothesis.

- Tests 1-16 below preserve the original historical development that led to the current frozen model. They are now placed at the front so the record reads continuously from Test 1 to Test 54.
- Current Test 17 is the trusted homogeneous bounce baseline. Tests 18-54 follow the restarted authoritative sequence used in current work.
- Historical Tests 18-22 from the older master ledger are preserved separately in Appendix A so they are not silently deleted or confused with current Tests 18-22.
- The decisive correction in Test 52 was the outer parent boundary: the positive-energy parent minimum supports an expanding vacuum branch with $H_{\text{parent}} = 0.2497149524$, not a static $W=0$ branch.
- The work still lacks dimensional calibration, observational fitting, a fundamental derivation of F , f and V , and a production-quality long-time strongly-hyperbolic numerical-relativity evolution.

2. Unified theory statement and frozen Trial Model 1

KRCM = Kinetic Recoil Cosmological Model. CBHO = Cosmic Black Hole Object. The working proposal is that the observable universe may be associated with a cosmic-scale gravitating object in a larger external setting, and that sufficiently strong curvature can drive a coupled two-scalar sector across a vacuum barrier. Depending on the global solution, a daughter/lobe region may remain connected through a throat, act as a temporary mass-transfer region, or potentially detach. Negative mass and white holes are not required.

- CBHO is not assumed to be an ordinary stationary Schwarzschild or Kerr black hole.
- $A \rightarrow B \rightarrow C$ labels recursive stages and does not imply that A is the first universe.
- No hard density cutoff, forced Hubble-sign reversal, LQG multiplier, phantom field, manually inserted daughter vacuum, or artificial repulsive force is permitted in the frozen model.
- Wormhole-like topology is optional and must emerge from a solution rather than be inserted by hand.

Frozen candidate action

$$S = \text{integral} \sqrt{-g} \left[F(h, \phi) R - \frac{1}{2} (dh)^2 - \frac{1}{2} (d\phi)^2 - V(h, \phi) + f(h, \phi) G \right] d^4x$$

$$F(h, \phi) = 1 + 0.08 h^2 + 0.06 \phi^2$$

$$f(h, \phi) = 0.82 h + 0.68 \phi$$

$$V = 0.300 + 0.25(h^2-1)^2 + 0.20(\phi^2-1)^2 + 0.09(h-\phi)^2 - 0.070 h - 0.055 \phi$$

$$G = R_{abcd} R^{abcd} - 4 R_{ab} R^{ab} + R^2$$

Current refined stationary points

Point	h	phi	V
Parent minimum	-0.9630260602	-0.9635971213	0.4227485835
Lowest parent-basin saddle	-0.8379303388	+0.1598286169	0.6515560938
Daughter minimum	+1.0332773412	+1.0327977708	0.1729002789
Central maximum	-0.0707416557	-0.0685881143	0.7543514440

Important correction: the old point near (-0.07,-0.069) is the central maximum, not the lowest parent-basin escape saddle. The current parent-to-daughter minimax barrier is V approximately 0.651556 through the saddle near (-0.838,+0.160).

Status labels

PASS: The stated gate passed inside the stated model, equations, and tested parameter range.

CONDITIONAL PASS: A meaningful gate passed, but the calculation remains reduced, local, proxy-based, short-time, or otherwise incomplete.

PARTIAL / MIXED: One sub-gate passed while another remained unresolved or failed.

FAIL: The tested construction or mechanism failed. A reduced-model failure is not automatically a theory-wide failure.

UNRESOLVED: No trustworthy physical pass/fail is available because the required formulation or equations remain incomplete.

3. Unified authoritative test sequence - Tests 1 through 54

This is the continuous sequence to use for public discussion. Tests 1-16 are the preserved historical development. Test 17 onward is the current authoritative restarted sequence. The older historical items once numbered 18-22 are preserved later in Appendix A to avoid a numbering collision.

1. Thin-wall bubble barrier removal - PASS / reproduced. Curvature-dependent wall modification erases the forbidden radial interval at $\alpha=0.20$ in the squared junction equation; barrier closes near $\alpha=0.151861$. Does not establish nucleation or pinch-off.

2. Static backreaction - REDUCED RESULT. Self-consistent backreaction moves fields from the fixed-background relaxation ($\sim 1.96, 1.92$) to ($\sim 0.54, 0.48$), demonstrating substantial feedback but not a complete daughter transition.

3. Horizon-penetrating Einstein evolution - PARTIAL / DIAGNOSTIC. Painleve-Gullstrand evolution crosses $\beta=1$ at $t\sim 0.164$. Later blowup is not accepted as a physical singularity because the field implementation/formulation became unbounded.

4. Scale-invariance check - CONCEPTUAL CORRECTION. Vacuum GR rescaling does not determine full KRCM behavior once scalar potentials, vacuum energies, wall tension and higher-curvature couplings introduce scales.

5. Matter tests - CONCEPTUAL / MIXED. Ordinary positive density alone does not imply universal reversal or no reversal. Pressure, curvature, anisotropic stress, flux, vacuum energy and scalar stresses matter.

6. High-density / density-cutoff exploration - HISTORICAL ONLY. Earlier modified models bounced, but arbitrary density cutoffs, force multipliers and forced sign flips were removed from the frozen Trial Model 1.

7. Recoil A->B daughter transfer shell - PASS BENCHMARK. Ordinary-GR/Vaidya shell benchmark has a sharp transfer boundary near $k_D=2.217426$ and demonstrates outward reversal without negative gravity. It is not a daughter-spacetime proof.

8. Quantum tunneling / vacuum decay arithmetic - CORRECTED. Corrected action scaling gives required epsilon $\sim 56.16\%$ of v_H^4 for $S_0=10$, not 0.06% . No physical KRCM tunneling rate has yet been derived.

- 9. Compactness / extra-GR correction - CONCEPTUAL CORRECTION.** The relativistic shell already contains GR effects; an arbitrary extra GR force multiplier is not legitimate.
- 10. Fuzzball / high-density floor - HISTORICAL ONLY.** Mechanical absorbing/reflecting floor was exploratory and is not a direct string-theory fuzzball calculation.
- 11. Radial impacts - HISTORICAL / MODEL-SPECIFIC.** Packet collision toy model explored infall and impacts but was not a full self-gravitating fluid or Einstein-scalar evolution.
- 12. Off-axis scattering - CONDITIONAL TO TOY MODEL.** No ejection in the fixed elastic-scattering setup; result is specific to the imposed mechanical rules.
- 13. Gravitational-radiation interpretation - CONCEPTUAL CORRECTION.** Energy loss cannot be converted into a universal rule that every bound system must tighten; specific dynamics must be modeled.
- 14. Merger buildup / one-CBHO formation - CORE MECHANISM UNRESOLVED.** Black-hole mergers can build a dominant object, but global collection of an entire universe requires a cosmological evolution, not local binary-merger arguments.
- 15. Negative mass requirement - NOT REQUIRED.** Current KRCM requires neither negative mass nor a phantom field or white hole. Wormhole-like geometry remains optional.
- 16. Universe compactness / Schwarzschild comparison - ARITHMETIC PASS, INTERPRETATION OPEN.** Formal compactness ratios can exceed unity for cosmological spheres, but $C > 1$ alone does not prove the universe is an ordinary black hole.
- 17. Homogeneous frozen-model bounce - PASS.** Trusted baseline: bounce $t = 6.187733109924$; DOP853/Radau agree to $\sim 3.5e-12$; matched GR and GB-off controls do not bounce before safety stops.
- 18. Generalized KRCM junction / transfer branch - PASS / LATER REFINED.** Generalized boundary terms and scalar momentum junctions produce a retained positive branch. Later corrected sharp-interface work is carried by Test 20.
- 19. Full spherical Einstein + two-field bridge - UNRESOLVED.** Old free synchronous spherical formulation loses convergence before the trusted bounce. Substitute smeared-interface and forced-source attempts are invalid for the physical question.
- 20. Corrected ratio-1.5 expanding worldtube - PASS.** Corrected $M_{\text{outer}}/M_B = 1.5$ branch reverses outward near $\tau = 0.01172454$ and stays outward through 99% transfer.
- 21. Worldtube mass-ratio robustness - PASS.** 39/39 mass-ratio cases pass from 1.10 to 3.00; approach to trapping appears only at substantially larger ratios.
- 22. Transfer-law robustness - PASS.** 1,281/1,470 broad cases pass and 600/600 core cases pass.
- 23. Temporary lobe persistence / mass return - PASS.** At reversal 92.281% of shell mass remains; the lobe persists through depletion and 144/144 perturbations pass.
- 24. Mass return versus geometric closure - MIXED.** Mass return passes, but geometric reclosure fails because the phenomenological recoil law retains finite outward acceleration as shell mass vanishes.
- 25. Interface dissolution / material reabsorption - PASS.** Interface excess decays to negligible levels while material is reabsorbed; 144/144 perturbations pass.
- 26. Daughter-channel branching - PASS.** Simultaneous B accretion and outgoing daughter-channel energy is robust inside the active-density window.
- 27. Daughter-seed freeze-out - PASS.** The outgoing seed freezes after gate-off with negligible post-gate drift.
- 28. Causal escape / no-recapture proxy - PASS.** Outgoing causal proxy stays positive and untrapped across baseline and robustness scan.
- 29. Detached-seed gravitational isolation proxy - PASS.** Seed coupling and tidal influence collapse strongly after freeze-out; 120/120 cases pass.
- 30. Daughter-seed self-gravity / immediate non-collapse - PASS.** Equivalent seed compactness remains subcritical; 120/120 pass.
- 31. Vacuum pressure versus wall collapse - FAIL.** Reduced thin-wall balance requires critical radius ~ 14.103 versus seed radius ~ 0.147 .

- 32. Kinetic/flyover support versus wall collapse - FAIL.** Available seed energy is far below the reduced thin-wall survival requirement.
- 33. Thick-wall flyover / seed-energy transition - FAIL.** Available uniform kinetic density is $\sim 76.32\%$ of the corrected parent-basin barrier requirement; no transition in this gate.
- 34. CBHO curvature work + thick-wall transition - CONDITIONAL PASS.** Curvature driver gives convergent barrier crossing above a sufficiently broad source scale; reduced no-gravity geometry later collapses.
- 35. Local-FLRW backreaction / thick-wall survival - CONDITIONAL PASS.** With local gradient redshift/backreaction, transition converges around $t \sim 17.95$ and survives to $t=60$; curvature-off control does not transition.
- 36. Apparent horizon / throat separation - PARTIAL PASS.** Daughter becomes super-horizon/anti-trapped in the reduced local-FLRW geometry, but that ansatz cannot represent a throat.
- 37. Two-sided areal-radius neck existence - CONDITIONAL PASS.** A finite constraint-compatible neck with $R'=0$ and $R''>0$ exists in a reduced two-sided proper-radial construction.
- 38. Dynamical throat persistence / pinch-off - CONDITIONAL PASS.** Reduced throat remains expanding; 25/25 perturbations persist with no pinch-off.
- 39. Throat stress tensor / matter compatibility - FAIL IN MINIMAL GR.** Canonical radial scalars have the wrong anisotropy sign for the required throat; full nonminimal KRCM remained open.
- 40. Full FR+fG reduced throat anisotropy - FAIL.** Homogeneous-scalar/isotropic self-similar neck cannot sustain the condition $F+4f_{\ddot{}}=0$.
- 41. Anisotropic extrinsic-curvature local-neck rescue - CONDITIONAL PASS.** Exact frozen nonminimal local neck expands and isotropizes; 7/9 geometry perturbations pass.
- 42. Radial-gradient characteristic / hyperbolicity gate - CONDITIONAL PASS.** Four physical roots remain real at $dr=.04,.02,.01$ and across all 7 passing Test-41 geometries.
- 43. Two-sided constraint-compatible radial slice - FAIL.** Constraints solve at machine precision, but the scalar-flat bridge does not connect cleanly to the ordinary parent side.
- 44. Curvature-solved parent connection - FAIL.** Allowing B'' to be determined by the constraints still produces extra radial turning surfaces with the tested W taper.
- 45. Joint $B(l)$, $W(l)$ two-point shooting - FAIL.** 2,904 primary shoots and refinements find no regular single-neck connection in the tested smooth K free-data families.
- 46. Reduced York/XCTS spectral free data - FAIL.** 65,536 spectral candidates plus optimization still miss the static-parent target; best branch remains anti-trapped.
- 47. Outer normal-parent local branch audit - CONDITIONAL PASS.** 2,739/2,772 scanned constant-parent states are normal-parent, hyperbolic, and constraint-compatible: no local algebraic no-go.
- 48. Global daughter-neck to normal-parent interpolation - FAIL.** 40,960 candidates reach some normal endpoints, but all develop another turning surface first.
- 49. Focusing / turning-point no-go gate - FAIL OF NO-GO.** 1,317 local $D'<0$ crossing states exist; 1,251 are hyperbolic. A local focusing theorem does not block the crossing.
- 50. Targeted hyperbolic crossing to parent - FAIL.** 57,344 targeted global feedback candidates produce zero valid single-throat static-parent branches.
- 51. Closed-parent equator reinterpretation - FAIL.** 28,672 candidates contain equator-like extrema, but converged parent-reaching branches retain multiple oscillatory extrema.
- 52. Correct expanding-parent boundary / single-throat rescue - PASS.** Using $H_{\text{parent}}=0.2497149524$ rather than $W=0$ produces a monotone $B'>=0$, constraint-compatible, physically hyperbolic single-throat slice.
- 53. Short-time nonlinear evolution - CONDITIONAL PASS.** To $t=0.06$ the throat expands $\sim 1.704\%$, stays a minimum, constraints converge $\sim$ second order, and sampled physical roots remain real.
- 54. Extended nonlinear evolution / throat bifurcation - CONDITIONAL PASS.** No immediate pinch-off. The center becomes a bulge and symmetric off-center throats form near $x \sim 0.47$ by $t=1.75$; late-time convergence remains incomplete.

4. Detailed record - Tests 1 through 16

Test 1 - Thin-wall bubble test

Status: PASS / REPRODUCED

Purpose: test whether a curvature-dependent wall modification can remove the forbidden radial region in a thin-wall junction model.

- $\rho_{\text{outer}}=1.0$, $\rho_{\text{inner}}=0.3$, $\sigma_0=0.15$; $\sigma_{\text{eff}}(R)=\sigma_0(1+\alpha/R^2)$.
- $\alpha=0$: turning radius $R \sim 0.337995177$.
- $\alpha=0.02$: roots ~ 0.019216939 and 0.344410989 .
- $\alpha=0.05$: roots ~ 0.049798897 and 0.344251767 .
- $\alpha=0.10$: roots ~ 0.109070045 and 0.325410277 .
- $\alpha=0.20$: no forbidden interval; minimum $R_{\text{dot}}^2 \sim +0.316993296$.
- Gap closes near $\alpha \sim 0.151861061570$.
- Limit: this does not establish a tunneling rate, physical nucleation, direction of motion, daughter pinch-off, or the actual Trial-Model-1 coupling.

Test 2 - Static backreaction test

Status: REPORTED REDUCED RESULT

Self-consistent field/curvature feedback materially changes the two-field relaxation compared with a fixed background.

- Fixed-background relaxation: $h \sim 1.96$, $\phi \sim 1.92$.
- Self-consistent backreaction: $h \sim 0.54$, $\phi \sim 0.48$.
- This is only a partial displacement and does not independently demonstrate a stable daughter vacuum or pinch-off.

Test 3 - Einstein evolution / horizon-penetrating test

Status: PARTIAL / DIAGNOSTIC

The areal-radius gauge failed numerically at a horizon. A Painleve-Gullstrand/horizon-penetrating formulation crossed $\beta=1$ at $t \sim 0.164$.

- Later unbounded behavior in the chosen field implementation is not treated as proof of a physical singularity.
- Scientific lesson: horizon penetration itself was numerically demonstrated; late-time formulation stability was not.

Test 4 - Scale-invariance check

Status: CONCEPTUAL CORRECTION

Vacuum Einstein equations admit rescaling, but full KRCM contains scalar-potential, vacuum-energy, wall-tension and higher-curvature scales. Vacuum-GR scaling therefore cannot by itself determine the complete KRCM dynamics.

Test 5 - Matter tests

Status: CONCEPTUAL / MIXED

Ordinary positive density alone does not produce a universal sign reversal and also does not rule out a bounce. Relevant quantities include pressure, anisotropic stress, flux, spatial curvature, vacuum energy and scalar stresses.

- Later Test 17 shows closed FLRW plus positive vacuum energy can bounce even in ordinary GR for suitable data, so "positive matter means bounce impossible" is not a valid general statement.

Test 6 - High-density / density-cutoff exploration

Status: HISTORICAL MECHANISM ONLY

Earlier modified models inserted high-density corrections and produced reversals/bounces. These were useful exploratory mechanisms but are not part of the current frozen model.

- An arbitrary force multiplier is not a derivation of loop quantum cosmology.

- A shift-function sign is not a gauge-invariant bounce criterion.
- No hard density cutoff, forced H flip, or LQG multiplier remains in Trial Model 1.

Test 7 - Recoil / A->B / daughter transfer shell

Status: PASS ORDINARY-GR BENCHMARK

Historical spherical radiating-dust shell with ingoing/outgoing Vaidya regions and Israel junction conditions demonstrated outward reversal without negative gravity.

- Original saved values: $R_0=3$, $V_0=-0.025$, $M_{B0}=0.08$, $M_{\text{outer}0}=1$, $q=0.5$, $\mu_0=1.1861934467462916$.
- Sharp TURN/NO-TURN boundary: $k_D=2.2174224853515625$ versus 2.2174301147460938 ; Radau confirmed.
- Largest audit residuals: angular Israel $2.69\text{e-}12$; normalized temporal junction $1.66\text{e-}12$; outer energy budget $1.34\text{e-}15$; local shell energy $5.61\text{e-}12$.
- At the x_2 turning point: attractive mass terms $=-0.082268204$, reaction into $B=+0.700225633$, daughter-flow reaction $=-0.421188235$, net acceleration $=+0.196769195$.
- The attractive gravitational terms never changed sign; outward motion did not require negative gravity.

Test 8 - Quantum tunneling / vacuum decay arithmetic

Status: CORRECTED

Historical thin-wall tunneling arithmetic was corrected.

- $S(R)=2 \pi^2 R^3 \sigma - 1/2 \pi^2 R^4 \epsilon$; $R_{\text{saddle}}=3 \sigma/\epsilon$; $S_0=27 \pi^2 \sigma^4/(2 \epsilon^3)$.
- With historical normalization $S_0=8.7\text{e}10$ at $\epsilon=1\text{e}6 \text{ GeV}^4$, $S_0=10$ requires $\epsilon \sim 2.056710116\text{e}9 \text{ GeV}^4$.
- $246^4 \sim 3.662186256\text{e}9 \text{ GeV}^4$, so the required ϵ is $\sim 56.1607\%$ of v_H^4 , not 0.06% .
- No physical KRCM tunneling/pinch-off rate has yet been calculated.

Test 9 - Compactness / extra-GR correction question

Status: CONCEPTUAL CORRECTION

The relativistic shell already contains mass, binding, Vaidya flux and Israel junction conditions. An arbitrary extra "GR force multiplier" cannot legitimately be applied on top; stronger compactness physics must come from a consistent new system of equations and initial data.

Test 10 - Fuzzball / high-density floor

Status: REPORTED HISTORICAL EXPLORATION ONLY

Absorbing/reflecting high-density floor models produced stopping/oscillation behavior in the historical session record. They were mechanical proxies, not direct string-theory fuzzball calculations, and the raw source was not preserved to the later verification standard.

Test 11 - Radial impacts

Status: HISTORICAL / MODEL-SPECIFIC

A packet/impact toy model explored radial infall and collisions. It tested the imposed collision rules and geometry, not a complete self-gravitating fluid or Einstein-scalar solution.

Test 12 - Off-axis scattering

Status: CONDITIONAL TO TOY MODEL

Historical elastic-scattering tests produced no ejection under the selected setup. A fixed elastic model cannot manufacture an unspecified extra energy reservoir; the result depends on the chosen energies, impact parameters and scattering prescription.

Test 13 - Gravitational radiation

Status: CONCEPTUAL CORRECTION

Binary inspiral through gravitational radiation is valid physics, but "energy loss always tightens every gravitationally bound system" is false as a universal rule. The response depends on the system and the way energy/mass is lost.

Test 14 - Merger buildup / one-CBHO formation

Status: CORE MECHANISM, GLOBAL DYNAMICS UNRESOLVED

The hypothesis allows black holes to merge/grow toward a dominant cosmic-scale object during contraction. Local mergers do not prove collection of an entire universe or a particular collapse duration; a global cosmological solution and initial distribution are required.

Test 15 - Negative mass

Status: NOT REQUIRED

Current KRCM does not require negative mass. Ordinary positive-density matter does not become negative gravitational mass merely because density rises.

- Negative mass: not required.
- White hole: not required.
- Wormhole: optional geometry only.
- Phantom field: not part of frozen Trial Model 1.

Test 16 - Universe compactness / Schwarzschild-radius comparison

Status: ARITHMETIC REPRODUCED; INTERPRETATION OPEN

A formal isolated-Schwarzschild comparison gives large compactness for an observable-universe-sized sphere, but an FLRW cosmology is not an isolated Schwarzschild object.

- Observable radius used historically: $4.39923966948 \times 10^{26}$ m (~ 46.5 Gly).
- $H_0 = 2.184285241 \times 10^{-18} \text{ s}^{-1}$; $\rho_{\text{crit}} = 8.532855 \times 10^{-27} \text{ kg/m}^3$.
- $M_{\text{total}} = 3.0430973936 \times 10^{54} \text{ kg}$; matter mass at $\Omega_m = 0.315$ is $9.5858 \times 10^{53} \text{ kg}$.
- Formal isolated Schwarzschild radius $= 4.5197 \times 10^{27} \text{ m}$; $R_s/R \sim 10.2738$.
- Separate 14.4-Gpc recheck: total compactness $= 10.481016$, matter-only $= 3.301520$, baryon-only $= 0.516811$.
- Compactness alone cannot distinguish an expanding from contracting FLRW sphere or establish that the universe is an ordinary black hole.

5. Detailed record - Tests 17 through 30

Test 17 - Trusted homogeneous frozen-model bounce

Status: PASS

Frozen Trial Model 1, closed homogeneous FLRW, $h = \phi = -1$, zero field velocities, dust/vacuum $= 1.0$, $\eta = 0.944$.

- $a_0 = 2.9196620900098154$, $H_0 = -0.083421006512061385$, $\rho_{d0} = 0.425$.
- Bounce time $= 6.187733109924$; min $a/a_0 = 0.828330313$; $H(t=30) = +0.249223$ and still expanding.
- Radau bounce $= 6.187733109920$, difference $\sim 3.5 \times 10^{-12}$.
- Matched GR control: no bounce before stop $t = 15.1691$. GB-off: no bounce before stop $t = 14.4086$.
- Scope: reduced homogeneous existence gate only; not daughter formation or full radial hyperbolicity.

Test 18 - Generalized KRCM A->B junction / transfer

Status: PASS / LATER REFINED

The generalized scalar-tensor plus Gauss-Bonnet boundary structure was set up with continuity of h , ϕ and their generalized momenta. A positive branch was retained and became the basis for the corrected sharp-interface work.

- Boundary action includes $2 F K$ plus the scalar-dependent Gauss-Bonnet boundary combination $4 f(2 G_{\text{bar}}{}_{ij} K^{ij} - J)$.
- Scalar momenta: $\Pi_h = -h_n + 0.32 h K + 3.28 Q$ and $\Pi_\phi = -\phi_n + 0.24 \phi K + 2.72 Q$, with $[h] = [\phi] = [\Pi_h] = [\Pi_\phi] = 0$.
- Important: the metric junction contribution from $f(h, \phi)G$ must be derived from the same scalar-dependent action; a constant-GB formula cannot simply be pasted in.
- Later ratio/mass normalization was corrected; use Test 20 values for the retained phenomenological worldtube.

Test 19 - Full spherical Einstein + two-field daughter-formation bridge

Status: UNRESOLVED

The physical target is a strongly-hyperbolic spherical 3+1 evolution of the metric, h , ϕ , Test-17 matter, the localized Test-18 interface, surface stress and energy/radial-momentum fluxes. No daughter vacuum may be inserted by hand.

- Required diagnostics include Hamiltonian/momentum constraints, field and metric residuals, Misner-Sharp mass, both null expansions, horizons, throats, interface variables, junction residuals, energy/momentum budgets and the physical principal symbol.
- The old free synchronous spherical solver loses constraint convergence before the trusted Test-17 bounce; that is a numerical-formulation failure, not a physical KRCM failure.
- Finite-width smeared-interface attempts were rejected because exterior mass and transfer-gate values changed strongly with resolution.

Test 20 - Corrected ratio-1.5 expanding bubble / worldtube

Status: PASS

The corrected sharp interface uses $M_{\text{outer}}/M_B=1.5$ rather than the obsolete 12.5 ratio.

- $R_0=0.6551156161993358$, $V_0=-0.025$, $M_{B0}=0.01746974976531562$, $M_{\text{outer}0}=0.026204624647973428$, $\mu_0=0.009038659013573232$.
- Retained positive generalized-junction branch: $R_{\text{ddot}0}=+3.636438397865514$.
- Outward reversal $\tau \sim 0.01172454151$; outward branch persists through 99% transfer.
- This remains a phenomenological ordinary-GR/Vaidya worldtube, not the full scalar-GB PDE.

Test 21 - Worldtube mass-ratio robustness

Status: PASS

39/39 cases pass for corrected M_{outer}/M_B ratios 1.10-3.00. DOP853/Radau reversal and transfer-stop times agree to very high precision. Trapping is only approached for ratios in the high teens, outside the retained baseline range.

Test 22 - Transfer-law robustness

Status: PASS

Across 1,470 combinations of transfer parameters, 1,281 broad cases pass; the defined 600-case core passes 600/600. Core maximum compactness remains approximately ≤ 0.080285 .

Test 23 - Temporary lobe persistence / mass return

Status: PASS

The lobe exists for a resolved finite interval while carrying mass and returning/redistributing it.

- At reversal 92.281% of shell mass remains.
- Shell-remaining milestones: 90% at $\tau \sim 0.01539$, 75% ~ 0.04231 , 50% ~ 0.10407 , 25% ~ 0.22202 , 10% ~ 0.41646 , 5% ~ 0.56783 , 1% ~ 0.91928 .
- Radius grows by $\sim 7.693x$ from reversal to the 1% stop; 144/144 perturbations pass.

Test 24 - Mass return versus geometric closure

Status: MIXED

Mass return succeeds, but geometric reclosure fails in the present phenomenological law.

- As $\mu \rightarrow 0$ the recoilB term remains finite because $\text{recoilB} = L_B \beta_o / \mu = k_B \beta_o$.
- At gate-off the baseline acceleration is still strongly outward, explaining why the branch does not geometrically close.

Test 25 - Interface dissolution / material reabsorption

Status: PASS

Evolution to $\mu/\mu_0=1e-8$ shows the interface excess becomes negligible while material is reabsorbed. Baseline stop $\tau \sim 3.9362$; interface excess compactness $\sim 8.28e-10$; 144/144 perturbations pass.

Test 26 - Daughter-channel branching

Status: PASS

The reduced law supports nonzero outgoing daughter-channel energy while B simultaneously accretes. Baseline delta $M_B/M_{\text{outer0}} \sim 17.34\%$; daughter-channel energy/ $M_{\text{outer0}} \sim 8.77\%$; 120/120 active perturbations pass.

Test 27 - Daughter-seed freeze-out

Status: PASS

The transfer gate shuts off around $\tau \sim 0.3220$ with $\sim 15.41\%$ shell remaining. After gate-off the daughter flux is zero and the emitted seed remains numerically frozen to extremely small drift. 120/120 robustness cases pass.

Test 28 - Daughter causal escape / no-recapture proxy

Status: PASS

The outgoing causal proxy $\beta_{\text{out}} + V$ remains positive and the branch remains untrapped in the reduced Vaidya picture; 120/120 cases pass. This is a causal-escape proxy, not a topology proof.

Test 29 - Daughter-seed gravitational-isolation proxy

Status: PASS

The seed gravitational-coupling and tidal proxies collapse strongly after freeze-out; baseline coupling reduction exceeds ~ 5.7 million by shell depletion. 120/120 cases pass.

Test 30 - Daughter-seed self-gravity / immediate non-collapse

Status: PASS

Equivalent uniform-sphere seed: $r_D \sim 0.14697$, Schwarzschild radius ~ 0.004598 , ratio $r_s/r_D \sim 0.0313$. 120/120 perturbations remain immediately subcritical.

6. Detailed record - Tests 31 through 42

Test 31 - Daughter-vacuum pressure versus wall-collapse survival

Status: FAIL

Reduced thin-wall balance gives wall tension ~ 1.76182 and critical radius ~ 14.10315 . The baseline seed radius ~ 0.14697 is roughly 96 times smaller. 0/120 robustness cases pass.

Test 32 - Kinetic/flyover support versus wall-collapse

Status: FAIL

Reduced thin-wall kinetic-support requirement $E_{\text{crit}} \sim 1467.86$ while baseline seed energy is ~ 0.47488 . The reduced wall-survival shortfall is overwhelming; 0/120 cases pass.

Test 33 - Thick-wall flyover / seed-energy transition

Status: FAIL

The corrected parent-to-daughter minimax barrier is set by the saddle $V \sim 0.651556$, not the central maximum.

- $D_{\text{seed}} \sim 0.002299$; seed radius ~ 0.14697 ; Delta $V \sim 0.226556$.
- Available uniform kinetic density ~ 0.172900 , 76.32% of the required barrier energy density.
- No transition is found even when concentrating the available energy into much smaller radii.

Test 34 - CBHO curvature work + thick-wall transition

Status: CONDITIONAL PASS

A historical reduced curvature driver can push the two-field profile across the barrier when the forcing region is broad enough.

- Initial compact source scale does not transition; an expanded source scale does.
- Critical source radius is around 3.151-3.152 in model units.
- Curvature-off control does not transition.
- The daughter later collapses in the flat/local scalar PDE, so this is a formation/barrier-crossing result only.

Test 35 - Local-FLRW backreaction / thick-wall survival

Status: **CONDITIONAL PASS**

Replacing the fixed-gradient term with local-FLRW redshifting, $a^{-2} \nabla^2$, allows the transitioned daughter to survive after the curvature source fades.

- Transition converges at $t \sim 17.95$ for $dr = .04, .02, .01$.
- No collapse through $t = 60$; comoving daughter core ~ 4.36 .
- Curvature source is effectively gone by $t \sim 18.58$.
- No-gradient-redshift control collapses near $t \sim 22.6249$; curvature-off control does not transition.
- Scope: reduced local-FLRW backreaction, not the full spherical FR+fG spacetime.

Test 36 - Apparent horizon / throat separation

Status: **PARTIAL PASS**

The Test-35 daughter crosses a cosmological apparent-horizon condition $H R = 1$ around $t \sim 17.044$ and becomes anti-trapped/super-horizon, not black-hole trapped. The imposed $R = a r$ geometry forces $R_r > 0$ and therefore cannot represent a throat.

Test 37 - Two-sided areal-radius neck / throat existence

Status: **CONDITIONAL PASS**

A proper-radial metric with $R'(0) = 0$ and $R''(0) > 0$ supports a finite constraint-compatible neck in a reduced construction.

- Representative $R_{\text{neck}} \sim 3.51$ and $R_{\text{daughter}} \sim 8.61$; $R_{\text{neck}}/R_{\text{daughter}} \sim 0.408$.
- Constraint residuals remain at machine precision and the tested flux ratio stays subluminal.
- This establishes a compatible two-sided initial geometry, not a full dynamical solution.

Test 38 - Dynamical throat persistence / pinch-off

Status: **CONDITIONAL PASS**

A reduced throat evolution remains expanding through $t = 60$. All 25 perturbations persist while frozen-w/dust controls pinch. This supports an optional persistent-neck branch but is not a topology proof.

Test 39 - Throat stress-tensor / matter compatibility

Status: **FAIL FOR MINIMAL CANONICAL SUPPORT**

For the self-similar scalar-flat throat, Einstein anisotropy requires $p_t - p_r = 3/(2R^2) > 0$, but canonical radial scalar gradients give $p_t - p_r = -(h_l^2 + \phi_l^2) \leq 0$. The signs oppose each other. Full nonminimal support remained unresolved at this stage.

Test 40 - Full FR+fG reduced throat anisotropy / local kinetic matrix

Status: **FAIL**

For a homogeneous-scalar/isotropic self-similar neck, exact frozen nonminimal anisotropy reduces to the sustained condition $F + 4 f_{\text{ddot}} = 0$. The Test-35 source-off history crosses that condition only briefly and does not sustain it.

Test 41 - Anisotropic extrinsic-curvature local-neck rescue

Status: **CONDITIONAL PASS**

Relaxing isotropic extrinsic curvature rescues a local exact nonminimal neck.

- Baseline $R_0 = 3.5086975324$, $W_0 = 0.2850060431$, $U_0 = -0.07167570185$.
- Over 20 model-time units the local neck expands strongly and shear decays toward isotropy.
- Constraint residuals are tiny; minimum F remains positive; min local acceleration-matrix $r_{\text{cond}} \sim 0.0953$.
- DOP853/Radau agree closely; 7/9 geometry perturbations pass.

Test 42 - Radial-gradient characteristic / hyperbolicity gate

Status: **CONDITIONAL PASS**

The actual Test-35 source-off scalar profile is inserted into a local frozen-coefficient spherical principal-symbol calculation around the Test-41 anisotropic neck.

- At $dr=.04,.02,.01$ the four nonzero physical roots are all real.
- All seven passing Test-41 geometry variants retain real physical roots.
- Artificially amplifying the whole scalar-gradient profile produces the first complex pair only near $\sim 4.657x$.
- Scope: local frozen principal symbol, not yet a global nonlinear radial evolution.

7. Detailed record - Tests 43 through 54

Test 43 - Two-sided constraint-compatible radial initial slice / hyperbolicity

Status: FAIL FOR SCALAR-FLAT BRIDGE

Full frozen Hamiltonian and momentum constraints solve to $\sim$ machine precision on the prescribed scalar-flat Test-37 bridge, but no ordinary parent-side connection is obtained. A small complex characteristic pair shrinks with refinement and remains unresolved in the continuum.

Test 44 - Curvature-solved two-sided parent connection

Status: FAIL

Dropping the scalar-flat geometry and allowing the momentum/constraint system to determine spatial curvature does not yield a regular single-neck parent connection for the tested prescribed W-taper families. Some branches reach $\theta_{\text{minus}} < 0$ but only after $B' < 0$ creates another radial turning surface.

Test 45 - Joint $B(l)$, $W(l)$ two-point shooting

Status: FAIL

Because the constraints are underdetermined, Test 45 prescribes smooth mean curvature $K(l)$ as gravitational free data and solves B and W jointly.

- Eight taper families, K_0 scan $[-2,4]$, three scalar resolutions: 2,904 primary shoots.
- No regular single-neck ordinary-parent solution in the tested family.
- Rare complete branches have large negative B' or miss the parent target.

Test 46 - Reduced York/XCTS-style spectral free-data boundary solve

Status: FAIL

Replacing a one-function taper with a flexible spectral $K(l)$ family still fails the static-parent target.

- 65,536 spectral candidates plus nonlinear optimization.
- Best optimized complete branch converges to $W(L) \sim 0.249735$ and $\theta_{\text{minus}} \sim +0.49936$, decisively anti-trapped under the imposed static-parent target.
- This rules out the tested reduced spectral family, not a full scalar-GB XCTS system.

Test 47 - Outer normal-parent branch / gauge-freedom audit

Status: CONDITIONAL PASS

A local analytic and numerical audit shows the anti-trapped outer state from Tests 43-46 is not forced by the frozen constraints.

- In the constant parent vacuum, $W=0$ and $B'>0$ gives $\theta_{\text{plus}}>0$, $\theta_{\text{minus}}<0$ with exact local constraints.
- 2,739 of 2,772 scanned local states are normal-parent, hyperbolic and exact-constraint compatible.
- Therefore the obstruction is global matching/free-data selection, not a local no-go.

Test 48 - Global interpolating daughter-neck to normal-parent slice

Status: FAIL

A broader global free-data search uses all seven passing Test-41 neck geometries, the actual Test-35 scalar wall, a smooth parent-vacuum completion and spectral $U(l)$ free data.

- 40,960 global candidates; 10,464 reach the outer boundary; 190 reach a normal-parent endpoint.
- 0/190 retain a single throat. Every normal endpoint first develops $B'<0$ somewhere.

Test 49 - Global focusing / turning-point no-go gate

Status: FAIL OF THE PROPOSED NO-GO

The question was whether the constraints force B' to reverse before θ_{minus} can cross zero. They do not.

- 131,072 exact crossing-surface states sampled.
- 1,317 have $B' > 0$, $\theta_{\text{minus}} = 0$ and the correct crossing direction $D' < 0$.
- 1,251/1,317 also pass the physical hyperbolicity gate.
- Thus no local focusing theorem explains the global failures.

Test 50 - Targeted hyperbolic crossing / global parent shooting

Status: FAIL

Using a specific Test-49 hyperbolic crossing state as an interior target still fails to connect the Test-41 neck globally while retaining a single throat.

- Direct optimized shoot misses the target and develops min $B' \sim -0.1507$.
- 57,344 additional targeted feedback branches across several geometrically constrained families yield 0 valid neck-to-static-parent branches.

Test 51 - Closed-parent equator compatibility / turning-surface classification

Status: FAIL

The extra $B' = 0$ surfaces were reinterpreted as possible equators of a closed FLRW parent, where $B' = 0$ and $B B'' + 1 = 0$.

- 28,672 candidates; 7,241 contain parent-zone extrema; 163 later enter an untrapped region.
- Best local closed-FLRW equator residual $|B B'' + 1| \sim 9.95e-5$.
- Fine-grid survivors still contain 3-5 maxima with intervening minima. No clean single-equator parent attachment is found.

Test 52 - Expanding-parent hyperbolic single-throat rescue

Status: PASS

The key correction is that the frozen parent minimum has positive vacuum energy, so its flat asymptotic branch is expanding rather than static.

- $F_{\text{parent}} \sim 1.1299047001$, $V_{\text{parent}} \sim 0.4227485835$.
- $H_{\text{parent}} = \sqrt{V_{\text{parent}} / (6 F_{\text{parent}})} = 0.24971495239871697$.
- Using $W \rightarrow H_{\text{parent}}$ and $U \rightarrow H_{\text{parent}}$ instead of $W \rightarrow 0/K > 0$ yields a monotone $B' \geq 0$ single-throat slice.
- Hamiltonian and momentum residuals are at machine precision by construction.
- The physical characteristic audit remains real across roughly 8,000 radial points per scalar resolution; min principal $r_{\text{cond}} \sim 0.059$ and max eigenvector condition ~ 5.92 .
- Outer solution asymptotes toward $W, U \rightarrow H_{\text{parent}}$ and $B' \rightarrow 1$ as the parent buffer is extended.

Test 53 - Short-time nonlinear evolution of Test-52 slice

Status: CONDITIONAL PASS

This is the first genuine nonlinear method-of-lines evolution of the Test-52 global slice using the full frozen radial residual system E_{xx} , E_{θ} , E_h and E_{ϕ} .

- Evolved to $t = 0.06$ at four radial resolutions.
- Finest throat radius: $3.5086975 \rightarrow 3.5684822$, growth $\sim 1.704\%$.
- Throat remains a minimum: finest minimum $B_{xx}(0) \sim 0.0040925 > 0$.
- F remains positive; evolution-matrix r_{cond} remains ~ 0.059 ; sampled final physical roots remain real.
- Observed absolute-constraint convergence order between the two finest runs: ~ 1.99 for E_{tt} and ~ 1.97 for E_{tx} .

Test 54 - Extended nonlinear evolution / throat bifurcation

Status: CONDITIONAL PASS

Extending the same evolution changes the geometry without immediate pinch-off.

- Central tracked $B \sim 3.80612$ at $t = .30$, ~ 4.11219 at $t = .60$, ~ 4.82820 at $t = 1.20$.

- Central B_{xx} crosses through zero around $t \sim 1.5-1.7$ depending on resolution: the reflection center changes from a local minimum to a local maximum.
- At $t=1.75$, better resolutions find a positive-side off-center throat at $x \sim 0.47070-0.47406$, $B \sim 5.65575$, $B_t \sim 1.6212 > 0$; symmetry implies a matching throat on the other side.
- Two better audited resolutions remain physically hyperbolic at $t=1.75$; the coarsest run shows a small complex pair. Late-time convergence therefore remains incomplete.

8. Other retained numerical and conceptual work

Pre-restart reduced-model results

- Frozen two-field robustness: 100/100 reported PASS.
- Zero-curvature control: no transition, PASS control.
- Historical curvature-amplitude threshold: $A_{\text{critical}} \sim 1.056$; frozen run $A=2.4$ was safely above threshold.
- Broad historical parameter scan: 119/500 reported successful reduced homogeneous parameter sets; full parameter archive was not retained to the later verification standard.
- Older thin-wall gravity model: positive-mass supercritical suite 100/100 expanded for initial radii 1.4-1.9 through $R=100$; subcritical controls 100/100 collapsed for initial radii .5-1.2. These thresholds belong to an older potential and must not be imported into current Trial Model 1.
- Older smooth Einstein initial-data constraint test found a horizon threshold $R_0 \sim 1.38351411$, about 2.94% above the older thin-wall threshold. This is not a current-model critical radius.
- Older spatial two-field bubble without local gravity reached maximum radius ~ 3.8 and collapsed at all three resolutions, establishing that local gravitational backreaction matters.
- Historical dynamic CBHO-trigger suite: 10/10 perturbed merger/contraction runs reportedly reached a persistent daughter-state transition in a reduced dynamic-backreaction model; the surviving record does not fully define the curvature normalization.

Thermal / CMB / BBN state

Historical work records reported a CMB gate as PASS and a BBN calculation involving helium and deuterium. Those outputs predate the current global Test-52 branch and have not been recovered/re-audited to the same standard. They should be treated as historical reported results until rerun with the current frozen model and global branch.

Higher-dimensional daughter architecture

A speculative extension allows developed daughters to separate along an additional spatial dimension or be divided by a higher-dimensional bulk/domain wall. This has not been solved. A credible version would require explicit 4+1D or higher equations, bulk/domain-wall stress-energy, junction conditions, stability and causal structure.

Observational ideas not yet verified predictions

- CMB structure
- Cold Spot / Eridanus region
- cosmic void boundaries and void lensing
- large-scale structure
- time-variable cosmic acceleration
- gravitational-wave signatures
- primordial black holes
- apparently old early objects
- high-energy/gamma emission

None of these is currently a verified prediction of the frozen full KRCM model. They become predictions only after the model is dimensionally calibrated and evolved through a consistent observational history.

9. Failures, invalid approaches, and corrections that remain part of the record

- Spatial two-field bubble without local gravity: reaches ~ 3.8 then collapses.

- Earlier separate A/B geometry: 0/10 junction trajectory success.
- Earlier $\kappa=0.3$ wall-coupling branch: 0/10.
- Some wall/geometry branches require negative exterior mass and are rejected as physical positive-mass branches.
- Historical low-eta homogeneous points: no bounce / tensor-gradient instability.
- Historical pure-radiation midpoint: later recollapse and tensor-gradient failure.
- Old free synchronous spherical formulation: loses constraint convergence before the trusted bounce.
- Finite-width smeared-interface construction: grid-dependent exterior mass/transfer gate; invalid formulation for the intended sharp worldtube question.
- Tests 31-33: several simple wall-survival/seed-energy mechanisms fail quantitatively.
- Tests 39-40: minimal canonical support and homogeneous/isotropic nonminimal throat support fail.
- Tests 43-51: repeated global static-parent searches fail; Test 52 shows these failures were strongly tied to the wrong asymptotic parent branch rather than a proven theory-wide no-go.
- Do not reintroduce obsolete physical ratio 12.5; the corrected sharp worldtube ratio is 1.5.
- Do not import the older critical radius ~ 1.3835 into the current frozen Trial Model 1.
- Do not require a white hole, negative mass, daughter return, lobe overshoot, forced H sign change, density cutoff or manually inserted daughter vacuum.
- Do not claim hyperbolicity merely because a solver ran; the physical principal symbol must be checked.

10. Current scientific status and next gates

What is internally supported by the current calculations

- A reproducible frozen homogeneous KRCM-specific bounce baseline exists.
- A phenomenological transfer/lobe law can move and return mass without invoking negative gravity.
- Curvature work can drive the two-scalar field profile across the corrected parent-basin barrier in reduced spatial simulations.
- With local gradient redshift/backreaction, the daughter field state can survive after the curvature source turns off in the reduced local-FLRW model.
- The exact frozen nonminimal equations admit a locally supported anisotropic neck and real physical characteristic roots for the actual Test-35 radial gradients.
- A full radial initial slice exists that joins the retained daughter neck to the correct expanding parent vacuum while remaining monotone and hyperbolic (Test 52).
- That slice survives short-time nonlinear evolution and, over the tested longer interval, evolves into a double-throat/central-bulge geometry rather than immediate pinch-off.

What remains unproven or incomplete

- Long-time convergence of the Test-54 bifurcation and the eventual fate of the off-center throats.
- Gauge/formulation independence: reproduce the result in an independent strongly-hyperbolic formulation.
- Fundamental derivation of the chosen F , f and V functions from established particle physics or a specific UV-complete model.
- Dimensional calibration from model units to physical scales and a consistent external/global spacetime interpretation.
- Updated thermal, BBN, CMB, structure-formation and late-time acceleration calculations using the current Test-52 global branch.
- At least one quantitative falsifiable prediction that differs from standard Lambda-CDM/GR.

Recommended next numerical gates

- Test 55: track the two Test-54 off-center throats with finer radial resolution and a second gauge/formulation; determine whether separation and late-time hyperbolicity converge.
- Move to a production-quality strongly-hyperbolic long-time evolution with improved constraint handling.
- Track invariant Misner-Sharp mass, both null expansions, apparent-horizon worldtubes and throat worldtubes through the bifurcation.
- Dimensionally calibrate the current branch and rerun the thermal/CMB/BBN/late-acceleration bridge.
- Derive a clean observational falsifier and compute its allowed parameter range.

Appendix A. Historical numbering crosswalk for archived Tests 18-22

The older master ledger had already used labels 18-22 before the project restarted the authoritative sequence at current Test 17. To prevent silent deletion, those archived items are preserved here under their historical labels. They are not the same as current Tests 18-22.

Historical 18 - finite eta window - PASS reduced homogeneous eta-window gate. KRCM-specific interval approximately $0.941330192566 \leq \eta < 0.944888072968$. Five interior points bounce and remain expanding to $t=30$; monitored constraints tiny. Low-eta failures and negative tensor-gradient values are preserved.

Historical 19 - constraint-projected radial perturbation - PASS as projected weakly nonlinear radial subtest. $n=2$, amplitude= $1e-3$, $\delta\phi=-(0.82/0.68)\delta h$. $N=8/12$ and timestep checks reproduce a mean bounce near 6.20 with tiny projected constraint residuals. It is not free evolution or a strong-hyperbolicity proof.

Historical 20A - dust-loaded free spherical evolution - NUMERICAL-FORMULATION FAIL. Constraint failure occurs earlier with increasing resolution ($N*t_{fail}$ roughly constant), including homogeneous controls. Treated as a bad formulation rather than a physical KRCM failure.

Historical 20B - matter-load robustness - PASS reduced homogeneous robustness. At $\eta=.944$ the result depends on matter load. High-precision interior points for dust/vacuum ratios .75-1.50 bounce and remain expanding with tiny constraints and Radau agreement.

Historical 21 - dust + radiation mixture - PASS WITH BOUNDARY FAILURE. Radiation shares 0-75% retain successful branches; a pure-radiation midpoint later recollapses and develops tensor-gradient failure.

Historical 22 - high-radiation mechanism - PASS mechanism diagnostic. Radiation pressure itself does not immediately kill the bounce. A nearby pure-radiation upper branch survives to $t=30$ while a lower branch later recollapses; failure is tied to later high-curvature scalar excursion rather than $\rho-3p=0$ at first bounce.

Appendix B. Public critique / Reddit posting guidance

For public critique, present the work as a speculative numerical hypothesis and ask reviewers to find the first incorrect equation, boundary condition, numerical assumption or missing invariant. Do not lead with a claim that the cosmology is proven.

Suggested title: Here is a hypothesis: A curvature-coupled two-scalar Gauss-Bonnet model can support a persistent daughter-region throat

Suggested framing: I am looking for technical criticism, not validation. The strongest current result is Test 52: an exact-constraint single-throat initial slice joining the retained daughter-neck state to the expanding parent vacuum, followed by short-time nonlinear survival in Test 53 and a tentative double-throat bifurcation in Test 54. I would specifically like criticism of the action variation/boundary treatment, the legitimacy of the free-data construction, the physical principal-symbol analysis, and whether the late-time throat bifurcation is likely to be a gauge or numerical artifact.

- State clearly that present simulations are in dimensionless model units unless explicitly calibrated.
- Disclose significant AI assistance for algebra/code/numerical workflow.
- Use GitHub or Zenodo for source code/raw JSON/data rather than linking an LLM chat or personal cloud drive.
- Post the concise critique brief first and link this complete record as supporting material.

Current bottom line

The frozen KRCM model has passed increasingly strong internal existence, constraint and short-time evolution gates, culminating in a monotone expanding-parent single-throat slice and a tentative nonlinear double-throat bifurcation. It remains a speculative numerical research program requiring independent derivation, replication, dimensional calibration and observational falsification.